# Journal of Graduate Medical Education


## Reliability of Large Language Models in Academic Research Evaluation: A Multi-Rater Comparison with Human Experts

--Manuscript Draft--


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| Manuscript Number: | JGME-D-26-00586 |
| Full Title: | Reliability of Large Language Models in Academic Research Evaluation: A Multi-Rater Comparison with Human Experts |
| Short Title: | Large Language Models in Academic Research Evaluation |
| Article Type: | Original Research |
| Keywords: | Artificial Intelligence; Education, Medical; Pathology; Reproducibility of Results |


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| Abstract: | Background: Large language models (LLMs) are increasingly being explored in academic medicine, but their reliability for substantive research evaluation remains uncertain.<br>Objective: This study assessed the performance of three LLMs in evaluating pathology research posters compared with a human faculty expert.<br>Methods: In this 2026 study, one pathology faculty member and three LLMs (ChatGPT-5.2, Gemini-3, and Claude Sonnet-4.5) independently evaluated 90 de-identified posters using a standardized rubric. The rubric included a modified poster type classification and three ordinal domains: methods strength, conclusion validity, and writing quality, each scored on a 1–5 scale. Inter-rater reliability was assessed using Krippendorff's α, Gwet's AC2, and weighted kappa.<br>Results: Poster type classification showed perfect agreement across raters. In contrast, agreement for ordinal domains was limited. Four-way exact agreement ranged from 0% for total score to 24.4% for conclusion validity, while mean pairwise agreement ranged from 23.1% to 57.8%. Weighted kappa values were low across domains, ranging from 0.058 to 0.226, with relatively higher agreement for writing quality. Agreement among LLMs consistently exceeded agreement between LLMs and the human evaluator. Scoring patterns also differed: ChatGPT assigned perfect scores across all three ordinal domains to 12.22% of posters, compared with 5.56% for Claude, 4.44% for Gemini, and none by the human evaluator.<br>Conclusion: LLMs showed strong reliability for categorical classification but limited consistency in ordinal research evaluation. These findings support their use as adjunctive tools rather than replacements for expert human reviewers. |

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May 29, 2026

Dear Editor-in-Chief,

We are pleased to submit our manuscript entitled “Reliability of Large Language Models in Academic Research Evaluation: A Multi-Rater Comparison with Human Experts” for consideration for publication in the Journal of Graduate Medical Education. This work represents a prospective inter-rater reliability study conducted at the University of Central Florida College of Medicine and the UCF Institute for Artificial Intelligence, with direct relevance to scholarly activity assessment and research mentorship in graduate medical education (GME).

Scholarly activity is a cornerstone of ACGME-accredited training programs, and program directors, faculty mentors, and assessment committees routinely evaluate resident and fellow research presentations (including poster projects) as part of milestone-based competency assessment. As artificial intelligence tools become increasingly accessible to educators, there is growing interest in whether large language models (LLMs) can assist with this evaluative workload. Yet rigorous evidence on their reliability as research evaluators remains scarce. Our study directly addresses this gap. We examined how three LLMs (ChatGPT, Gemini, and Claude Sonnet) performed in evaluating ninety de-identified pathology student/resident research posters against a standardized rubric, and we benchmarked their performance against an experienced faculty rater using multiple inter-rater reliability indices (Krippendorff’s α, Gwet’s AC2, and weighted κ).

Our findings carry important implications for GME programs considering AI-assisted evaluation of trainee scholarly work: (1) All raters achieved perfect agreement on categorical poster-type classification, suggesting LLMs can reliably handle discrete categorization tasks; (2) Ordinal domain scoring agreement was uniformly low across all rater pairs, with pairwise weighted κ values ranging from 0.058 to 0.226; (3) AI–AI pairs consistently outperformed human–AI pairs, suggesting systematic divergence in evaluative standards between human and artificial raters; and (4) Scoring inflation varied meaningfully across models, with ChatGPT assigning perfect scores to 12.22% of projects versus 0% for faculty. Together, these findings caution against deploying LLMs as autonomous evaluators of trainee research and argue for their use as adjunct tools that augment—rather than replace—expert faculty review.

We believe this manuscript makes a timely and substantive contribution to the literature on AI in medical education. It provides actionable guidance for program directors, faculty evaluators, and GME leaders considering LLM-assisted scholarly activity workflows, and it highlights the need for improved rubric design and standardized prompting protocols before such tools can be responsibly integrated into trainee assessment.

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This manuscript has not been published elsewhere and is not under consideration by any other journal. All authors have reviewed and approved the final version. There are no competing interests or external funding sources to declare. The authors disclose that large language model (LLM) assistance was used for text editing; all intellectual content, data, and conclusions are the sole responsibility of the authors. The authors' contributions are as follows: Hatem Kaseb and Pegah Khosravi designed the study. Margaret-Anne Huntley and Hatem Kaseb collected the data and wrote the manuscript. Sai Karthik Nallamothu, Pegah Khosravi, and Hatem Kaseb performed and reviewed the statistical analyses. All authors reviewed and approved the final draft.

We thank you for considering our work and look forward to your response. Please do not hesitate to contact us with any questions.

Sincerely,

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# Reliability of Large Language Models in Academic Research Evaluation: A Multi-Rater Comparison with Human Experts

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**Funding:** No external funding was received.

**Disclaimers:** None.

**Previous Presentations:** This work has not been previously presented.

**Data Availability:** Data are available upon reasonable request from the corresponding author.

**Competing Interests and Funding:** None**.**

**Abstract word count: 232 words**

**Manuscript word count: 2508 words**

**Contributions:** Hatem Kaseb and Pegah Khosravi designed the study. Margaret-Anne Huntley and Hatem Kaseb collected the data and wrote the manuscript. Sai Karthik Nallamothu, Pegah Khosravi, and Hatem Kaseb performed/reviewed the statistics. All the authors reviewed and edited the final draft.

**Acknowledgment:** The authors disclose the use of a Claude Sonnet 4.6 LLM to assist with language clarity, grammatical refinement and some figure design during manuscript preparation. The authors retain full responsibility for the content and interpretation of the work. Biostatistical consultation for this study was provided by Xiang Zhu, PhD, at the UCF College of Medicine.

**Abstract**

**Background:** Large language models (LLMs) are increasingly being explored in academic medicine, but their reliability for substantive research evaluation remains uncertain.

**Objective:** This study assessed the performance of three LLMs in evaluating pathology research posters compared with a human faculty expert.

**Methods:** In this 2026 study, one pathology faculty member and three LLMs (ChatGPT-5.2, Gemini-3, and Claude Sonnet-4.5) independently evaluated 90 de-identified posters using a standardized rubric. The rubric included a modified poster type classification and three ordinal domains: methods strength, conclusion validity, and writing quality, each scored on a 1–5 scale. Inter-rater reliability was assessed using Krippendorff's α, Gwet's AC2, and weighted kappa.

**Results:** Poster type classification showed perfect agreement across raters. In contrast, agreement for ordinal domains was limited. Four-way exact agreement ranged from 0% for total score to 24.4% for conclusion validity, while mean pairwise agreement ranged from 23.1% to 57.8%. Weighted kappa values were low across domains, ranging from 0.058 to 0.226, with relatively higher agreement for writing quality. Agreement among LLMs consistently exceeded agreement between LLMs and the human evaluator. Scoring patterns also differed: ChatGPT assigned perfect scores across all three ordinal domains to 12.22% of posters, compared with 5.56% for Claude, 4.44% for Gemini, and none by the human evaluator.

**Conclusion:** LLMs showed strong reliability for categorical classification but limited consistency in ordinal research evaluation. These findings support their use as adjunctive tools rather than replacements for expert human reviewers.

**Keywords**

Artificial Intelligence; Education, Medical; Pathology; Reproducibility of Results

## Introduction

Large language models (LLMs) have emerged as influential tools in healthcare and academic medicine, with the ability to process large volumes of text, generate summaries, and support a range of language-based tasks that may reduce administrative burden and assist scholarly work [1,2]. Built on deep learning and natural language processing architectures, LLMs have demonstrated strong performance in tasks such as summarization, content generation, and contextual reasoning [1,3]. Consequently, they are being increasingly investigated for use in medical education, research support, and academic writing processes, such as manuscript drafting, peer review facilitation, and research assessment [3–5].

Even with their growing capabilities, LLMs are being adopted in academic medicine at a measured pace, largely because of concerns about reliability, transparency, and academic integrity. A key limitation is the tendency of LLMs to produce incorrect or fabricated information, often referred to as “hallucinations” which can include inaccurate references, misleading statements, or incorrect clinical reasoning [2,3,6]. Additional challenges include variability in output consistency, difficulty keeping up with the latest literature, lack of access to subscription-only content and reduced accuracy in highly specialized domains [6–8]. Consequently, current evidence suggests that LLMs are best positioned as assistive tools that augment human expertise rather than replace it [1,3,7].

Prior studies have evaluated LLM performance in academic and clinical tasks, including standardized examinations, manuscript review, and guideline-based assessments [3, 7, 9]. While some studies report moderate agreement between LLMs and human evaluators in structured settings, their reliability in more complex subspecialty areas and multidimensional evaluations remains uncertain [9,10]. Evaluating scientific work is usually complex and rarely straightforward; it draws not only on language comprehension, but also on domain expertise, methodological understanding, and contextual judgment, areas where LLMs can fall short when used in isolation.

Within medical education, the evaluation of posters is a complex process that integrates structured assessment criteria with elements of informed subjectivity. This process is typically guided by rubrics that meticulously assess various domains, including study design, methodological rigor, conclusion validity, and presentation quality. As the landscape of biomedical research evolves, it increasingly incorporates sophisticated experimental designs, molecular techniques, and intricate statistical analyses, which impose greater cognitive demands on reviewers. For example, in pathology, research often brings together diverse elements such as histopathology, immunohistochemistry, genomics, and bioinformatics. Interpreting this integration requires specialized expertise to ensure accurate evaluation, given the inherent complexity of the data. In this context, LLMs may serve as valuable adjuncts, offering support through structured, rubric-based assessments. They are particularly well suited to evaluating clearly defined elements such as language quality, adherence to formatting guidelines, and basic structural components, thereby helping to streamline aspects of the review process.

As we navigate this intricate landscape, it becomes clear that the role of LLMs is not to replace human evaluators but to enhance their capabilities. By providing consistent and objective assessments for certain aspects of poster evaluations, LLMs can help alleviate some of the cognitive load on reviewers, allowing them to focus on the more complex elements of research evaluation that require human insight and expertise. Thus, the integration of LLMs into the evaluation process represents a promising avenue for improving the efficiency and effectiveness of research assessments in medical education [7,11]. In recent years, the excitement surrounding AI-assisted evaluation has surged, yet a significant gap persists in our understanding of how well LLMs can mimic human judgment in academic assessments. While many studies have explored the capabilities of individual models or specific tasks, they often overlook the broader picture. There is a lack of comprehensive research that systematically compares various LLMs against human expert evaluations, particularly using structured rubrics and diverse reliability metrics. This gap calls for a more profound investigation into the reliability of LLMs across multiple evaluators and models in real-world academic settings, paving the way for future studies to explore their potential in enhancing educational assessments [12,13].

Addressing a significant gap in the literature, this study utilizes a multi-modal, multi-rater framework to evaluate the performance of three LLMs (ChatGPT-5.2, Gemini-3, and Claude Sonnet-4.5) in assessing pathology research posters. By benchmarking these models against a human faculty expert using a standardized rubric, the research examines consistency across categorical and ordinal domains. A core contribution of this work is its methodological rigor, featuring a systematic inter-rater reliability analysis of human-AI and AI-AI pairs within an authentic academic environment. Ultimately, this study aims to quantify the efficacy of LLMs in structured scholarly assessment, defining their current utility and potential to enhance the evaluation of academic research.

## Methods

### Study Design and Context

This study employed a structured, multi-stage evaluation design conducted in 2026 to assess inter-rater reliability between a human faculty expert and three large language models (LLMs) in scoring de-identified student research posters (Figure 1). The four-stage pipeline encompassed poster collection and de-identification, rubric development and application, independent scoring by each rater, and statistical computation of agreement indices across categorical and ordinal domains.

### Sample and Evaluation Process

A total of 90 de-identified research posters were independently evaluated by one human faculty rater and three LLMs (ChatGPT-5.2, Gemini-3, and Claude Sonnet-4.5), all blinded to one another's assessments. Scoring utilized a standardized four-domain rubric: poster type (categorical) and three ordinal quality domains (method strength, conclusion validity, and quality of writing), each rated on a 5-point scale with no half-point increments permitted (Supplemental Table 1-2). Poster type was assigned fixed values reflecting study design complexity, with basic research receiving the highest score. Given the single human rater, analyses focused on concordance between each LLM and this human reference standard rather than traditional multi-human inter-rater reliability.

## LLM Configuration and Prompting Protocol

All three LLMs were accessed via their respective web-based interfaces during February–March 2026 under default system configurations, without API access or parameter modification. Each model evaluated all posters independently in a single pass (Supplemental Table 1-2) using a uniform, standardized prompt instructing the model to adopt the role of an expert academic physician and score strictly according to the rubric, without inferring beyond the provided content (Supplemental Table 1-2).

## Outcome Measures and Statistical Analysis

Descriptive statistics were utilized to summarize scoring distributions and perfect score rates by rater. Agreement was quantified using four-way exact agreement and mean pairwise exact agreement across all six rater dyads. Chance-corrected reliability was assessed using Krippendorff's α (nominal metric for poster type; ordinal distance weighting for quality domains) and Gwet's AC1/AC2 coefficients. Quadratic-weighted Cohen's κ was calculated for each rater pair to examine dyadic agreement patterns, with benchmarks ranging from poor (< 0.00) to almost perfect (> 0.80) [14]. A composite total score (range: 6–20) was computed by summing all four domain scores, with the same indices applied to this aggregate measure. Scoring stringency was characterized by calculating perfect score frequency per evaluator, and convergent validity was assessed by cross-tabulating each rater's top three highest-scoring projects.

## Ethical Considerations

The study was determined exempt from IRB human subjects requirements (exemption number: STUDY00009045), as all data were fully de-identified and derived from routine educational evaluation activities.

## Results

### Descriptive Characteristics and Evaluation Efficiency

Ninety research posters were analyzed, with case reports comprising the majority (78.9%), followed by case series (14.4%) and original research studies (6.7%). The distribution across submission types was statistically significant ($p < 0.001$) (Supplemental figure 1). Regarding evaluation efficiency, the faculty reviewer spent approximately 5 minutes per poster (~7.5 hours total), whereas each LLM completed evaluations in under one minute per poster (~1.5 hours total).

### Agreement trends across raters and Domains

Poster-type classification achieved perfect agreement across all four raters, with four-way exact agreement, mean pairwise exact agreement, Krippendorff's α, and Gwet's AC2 all equal to 1.000. Agreement was considerably lower across the three ordinal quality domains (Figure 2 and Supplemental table 3). For Methods Strength, four-way exact agreement was 14.4% and mean pairwise agreement was 52.2%; Krippendorff's α indicated slight reliability (0.037), while Gwet's AC2 suggested moderate-to-substantial agreement (0.612). Conclusion Validity showed the highest pairwise consistency among ordinal domains (57.8%), with a Krippendorff's α of 0.013 and AC2 of 0.644. Quality of Writing yielded the lowest AC2 (0.601), with four-way and pairwise exact agreement of 13.3% and 51.1%, respectively. The composite Total Score had 0% four-way exact agreement and 23.1% mean pairwise agreement, yet achieved the highest AC2 (0.702) and a Krippendorff's α of 0.125, attributable to ordinal-weighted partial credit across the broader scale.

Pairwise weighted κ values were in the slight-to-fair range across all ordinal domains (Figure 3; Supplemental Table 3). LLM–LLM pairs consistently outperformed faculty–LLM pairs, most notably in Quality of Writing, where LLM–LLM κw reached 0.226 (ChatGPT–Gemini) versus 0.149–0.168 for faculty–LLM pairs.

**Scoring Distributions and Top-Ranked Project Concordance**

Significant differences in scoring distributions were identified across raters ($\chi^2(3) = 13.13$, $p = 0.004$; Cramér's V = 0.191). The pathology faculty assigned no maximum composite scores (5-5-5), while ChatGPT awarded ceiling scores 12.22% of the time, followed by Claude (5.56%) and Gemini (4.44%), indicating a consistent tendency among LLMs toward score inflation relative to the human rater. Top-ranked project concordance was limited (Supplemental table 4). The only unanimously top-ranked project across faculty, Claude, and ChatGPT was Project 11. Three additional pairwise agreements were observed: Project 65 (faculty and Claude), Project 14 (Claude and Gemini), and Project 1 (ChatGPT and Gemini).

## Discussion

This study evaluated the reliability of three LLMs (ChatGPT-5.2, Gemini 3, and Claude Sonnet 4.5) relative to a human faculty rater in scoring 90 pathology research posters across four domains: Poster Type, Methods Strength, Conclusion Validity, and Quality of Writing. A striking divergence emerged between categorical and ordinal performance. All raters achieved perfect agreement on Poster Type classification, confirming that LLMs can reliably replicate expert categorical judgment when criteria are well-defined and mutually exclusive consistent with prior literature[11,15–18] and suggesting a role for these models in early-stage screening and administrative review[7,19, 20]. Ordinal scoring presented a contrasting perspective. Four-way exact agreement ranged from only 13.3% to 24.4% across the quality domains, with weighted kappa values falling in the slight-to-fair range for all rater pairs. The composite total score showed the highest chance-corrected agreement (Krippendorff's α = 0.125; Gwet's AC2 = 0.702), a pattern partly attributable to the stabilizing effect of the perfectly agreed-upon Poster Type classification. A notable discrepancy between the two agreement indices was noted. Krippendorff's α yielded very low values for ordinal domains (range: 0.013–0.062), while Gwet's AC2 produced substantially higher estimates (0.601–0.644), reaching the substantial range for Methods Strength (AC2 = 0.612). When scores cluster within a narrow range, Krippendorff's α is susceptible to artificial deflation through its chance-correction mechanism, a well-documented phenomenon known as the "prevalence paradox" [21–23]. Gwet's AC2 distributes its chance baseline more uniformly across categories and is less prone to this distortion. Even so, AC2 values in this study generally remained below the thresholds considered adequate for high-stakes autonomous evaluation, reinforcing the need for interpretive caution.

LLMs and the faculty reviewer differed not only in agreement levels but also in scoring behavior. Pairwise weighted κ analysis revealed that AI–AI dyads consistently demonstrated modestly higher concordance than human–AI dyads across all ordinal domains. For example, ChatGPT and Gemini achieved a κw of 0.226, while the faculty rater and Claude Sonnet achieved only 0.149 for Quality of Writing. This pattern suggests that LLMs share latent scoring tendencies, likely arising from similar training paradigms, that diverge from the implicit standards applied by experienced human reviewers. The raters also differed meaningfully in scoring behavior. The human faculty applied the most conservative ratings, awarding zero perfect composite scores, while ChatGPT assigned perfect scores (5–5–5) in 12.22% of cases, Claude in 5.56%, and Gemini in 4.44%. The faculty's conservatism likely reflects domain-specific expertise and implicit consideration of factors beyond the rubric, including study feasibility, clinical relevance, and risk of bias. This is illustrated by the Conclusion Validity domain: with 78.9% of posters being case reports, a design inherently limited in capacity to change clinical practice, awarding a top score is difficult to justify regardless of writing quality. LLMs, by contrast, appeared more responsive to narrative clarity and writing quality than to methodological rigor, a tendency consistent with findings by Latona et al., who reported that LLM-reviewed research is more likely to be accepted than when evaluated by human experts [14]. Collectively, these findings reveal a significant "judgment gap" between LLMs and human expert reviewers. Despite widespread variability, certain posters notably Projects 11, 65, 14, and 1 were consistently ranked among the top contenders across evaluators. Project 11 achieved the broadest consensus among the faculty, Claude Sonnet, and ChatGPT, suggesting that clear markers of scientific quality are recognizable to both human and machine evaluators. Notably, Projects 11 and 65 were also selected by the faculty among the best three posters on the day of the event, corroborating the real-world validity of this convergence.

Despite current limitations, LLMs offer meaningful support in targeted applications. They show promise in classifying research designs during early-stage review, reducing burden on expert reviewers and freeing faculty time for mentorship and case-based learning [20,24]. All three LLMs also generated detailed evaluative comments without explicit prompting, which could support scientific communication training and poster preparation [3,25,26]. However, increased reliance on LLM feedback risks reducing direct faculty contact, important for developing skills in visual design, figure labeling, and scientific argumentation. For high-stakes decisions such as summative assessment or grant review, LLMs are not yet appropriate as independent evaluators. A hybrid model in which LLMs assist rather than replace human reviewers reflects both the current evidence and emerging consensus in the field [11]. One practical approach involves calibrating models using expert-provided scores and rationales prior to deployment to reduce systematic scoring bias. LLMs currently lack the reliability needed for independent ordinal assessment of research quality. Until calibration improves, high-stakes academic evaluations should remain human-led, with LLMs serving as supportive tools that augment rather than replace expert judgment.

This study has several limitations. Only one human rater was used, preventing a robust human interrater reliability benchmark. The sample came from a single pathology educational meeting and consisted mainly of case reports, limiting generalizability. Because posters include visual elements such as figures, tables, and microscopic images, some LLMs may have been disadvantaged, particularly in pathology, where image-based interpretation is central and LLM performance on image-based assessments remains limited and in development [27]. Multi-point rubrics also show limited reliability even among experienced human reviewers [20,24,25]. Finally, rapid LLM development, opaque training processes, and rater agreement without proven validity limit reproducibility and interpretation.

Future research should focus on improving rubric design, since rubric quality can influence interrater reliability [20]. Greater specificity or simplification may enhance LLM scoring consistency; calibration exercises and hybrid human–LLM models could greatly improve alignment with faculty judgment. Studies should also assess longitudinal score stability, performance across disciplines, and the educational impact of AI-assisted feedback. Methods such as chain-of-thought prompting may clarify why LLMs diverge from experts, especially when judging conclusion validity. Overall, LLMs show promise for classification and structured evaluation, but high-stakes academic assessment should remain human-led, with LLMs serving as a complementary support tool.

Figure 1: Framework of the study showing the different stages, including dataset preparation and design (Stage 1); multi-rater assessment utilizing a standardized scoring rubric (Stage 2); and, finally, the inter-rater reliability (IRR) analysis.

The figure shows the three stages of the study. The objective was to evaluate the interrater reliability among one human expert and three large language model raters in the assessment of 90 research posters. In Stage 1, a curated dataset of projects (1–90) was collected and assembled. In Stage 2, all posters were independently scored using a standardized rubric across four domains: poster type (categorical) and methods strength, conclusion validity, and quality of writing (ordinal), with raters blinded to one another's scoring. In Stage 3, agreement was analyzed using different approaches, including raw agreement (four-way exact and mean pairwise), chance-corrected indices (Krippendorff's α and Gwet's AC2), and pairwise quadratic-weighted Cohen's κ, with established thresholds used to interpret agreement strength from slight to almost perfect.

Figure 2: Four-way versus mean pairwise exact agreement, ranges, stacked composition, and gaps across all evaluation domains. (E–F) The same comparisons for methods strength, conclusion validity, and quality of writing are shown as percentages for the ordinal κ summary.

(A) Bar plots demonstrate that the four-way exact agreement is consistently lower than mean pairwise agreement across all ordinal domains, with perfect agreement observed only for poster type. (B) Domain-specific agreement ranges highlight the spread between four-way and pairwise agreement, emphasizing greater divergence in subjective domains such as methods strength and quality of writing. (C) Stacked bar plots illustrate the contribution of four-way agreement and the additional agreement captured at the pairwise level, showing that a substantial proportion of agreement in ordinal domains is not reflected in full-consensus measures. (D) The agreement gap (pairwise minus four-way agreement) quantifies the magnitude of discordance across domains, demonstrating large gaps for methods strength and quality of writing, moderate gaps for conclusion validity and total score, and no gap for poster type. (E) Bar plots compare four-way exact agreement percentages against mean pairwise agreement, showing a consistent trend where pairwise agreement significantly exceeds group consensus. (F) Stacked bar plots visualize the additive nature of pairwise agreement over the foundational four-way exact agreement.

Figure 3. Comparative Pairwise Weighted Kappa Analysis Reveals Enhanced Internal Consensus Among Large Language Models Relative to Faculty Alignment

This figure illustrates the pairwise quadratic-weighted Cohen's kappa across three ordinal evaluation domains, comparing the performance of a faculty expert against three large language models (LLMs) and the LLMs against each other. **(A)** Bar plots show weighted kappa values for every possible rater pair, demonstrating that "Quality of Writing" consistently yields higher reliability than "Methods Strength" or "Conclusion Validity." **(B)** The weighted kappa heatmap provides a color-coded matrix of agreement, visually separating faculty-LLM pairs from LLM-LLM pairs. **(C)** Kappa trends highlight a distinct "reliability step-up" when comparing LLMs to one another versus comparing them to the human faculty expert. **(D)** Mean weighted kappa plots quantify this difference, showing that LLM-LLM pairs (average weighted kappa: 0.105–0.219) exhibit significantly higher internal consistency than faculty-LLM pairs (average weighted kappa: 0.065–0.157).

Figure 4. Percentage of Perfect Scores (5-5-5) Across Raters.

This bar chart illustrates the frequency of perfect scores (5–5–5) awarded across ordinal domains by different models and faculty (N=90). ChatGPT demonstrated the highest propensity for assigning maximum ratings (n = 11, 12.22%), followed by Claude Sonnet (n = 5, 5.56%) and Gemini (n = 4, 4.44%), whereas pathology faculty awarded zero perfect scores (0%). This distribution highlights a relative inflation in LLM-assigned ratings compared to the strict evaluative baseline maintained by expert human raters.

Supplemental Tables

Supplemental Table 1. System Prompt Used to Instruct Large Language Models for Standardized Evaluation of Pathology Research Posters

Supplemental Table 2. Research Poster Scoring Rubric: Domains of assessment included Poster type, Methods Strength, Conclusion Validity, and Quality of writing.

Supplemental Table 3: Comprehensive Analysis of Interrater Reliability Indices and Estimates.

Supplemental Table 4. Overview of Top-Ranked Models Across Projects Cross-Model Consensus on Best Papers Project Models Ranking in Top 3.

Supplemental Figures

Supplement Figure 1. Distribution of Research Poster Types Among Analyzed Projects (N = 90)

Bar chart depicting the distribution of poster types among the 90 evaluated research posters. Case reports comprised the majority of submissions (n = 71, 78.9%), followed by case series (n = 13, 14.4%) and original basic research studies (n = 6, 6.7%), reflecting a statistically significant predominance of case-based submissions within the cohort ($\chi^2(2) = 84.87$, $p < 0.05$). Poster type was utilized in the scoring rubric: case reports were assigned a baseline score of 3 points, case series and other observational studies received 4 points, and original basic research was awarded the maximum of 5 points. This tiered scoring system recognized the greater longitudinal time investment and methodological rigor inherent in more complex study designs.

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Figure 1

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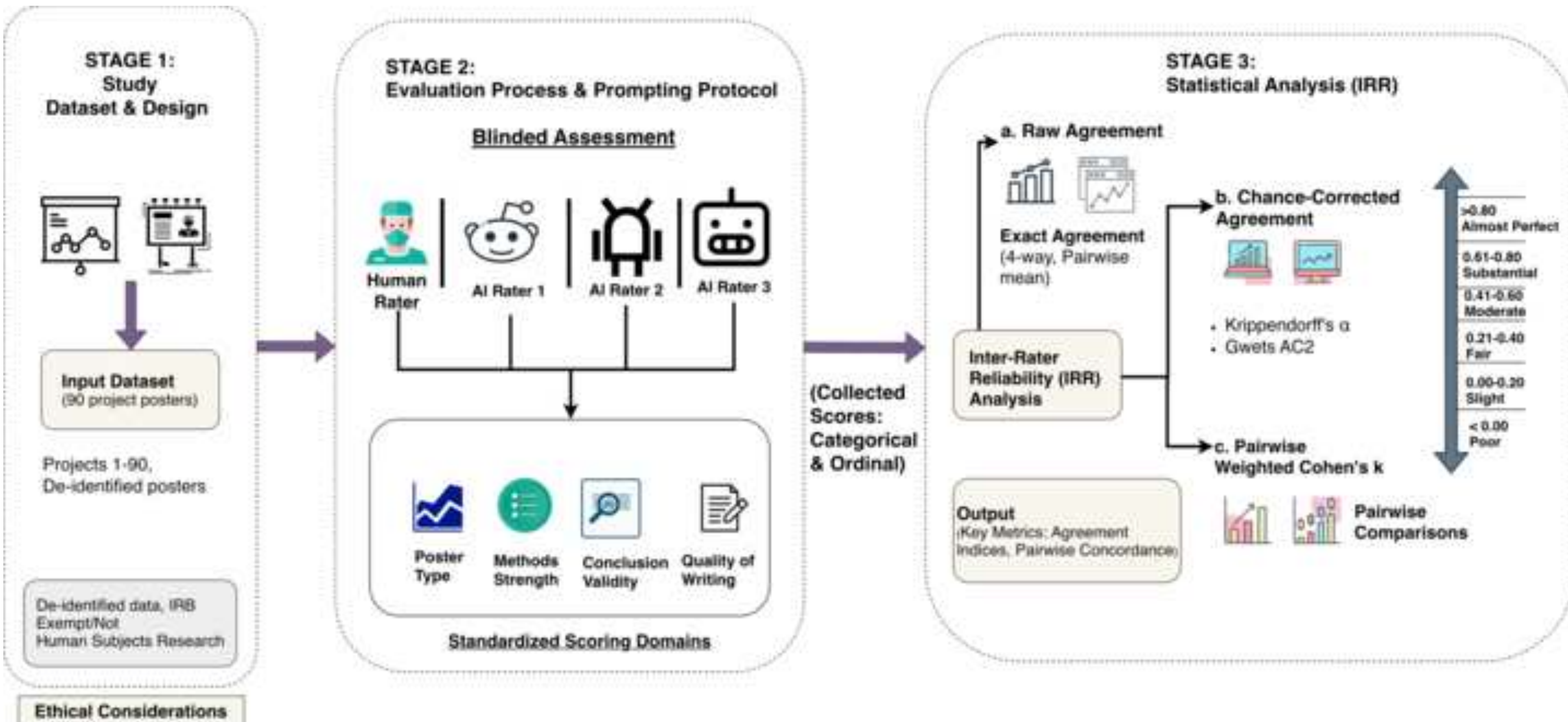

Figure 2

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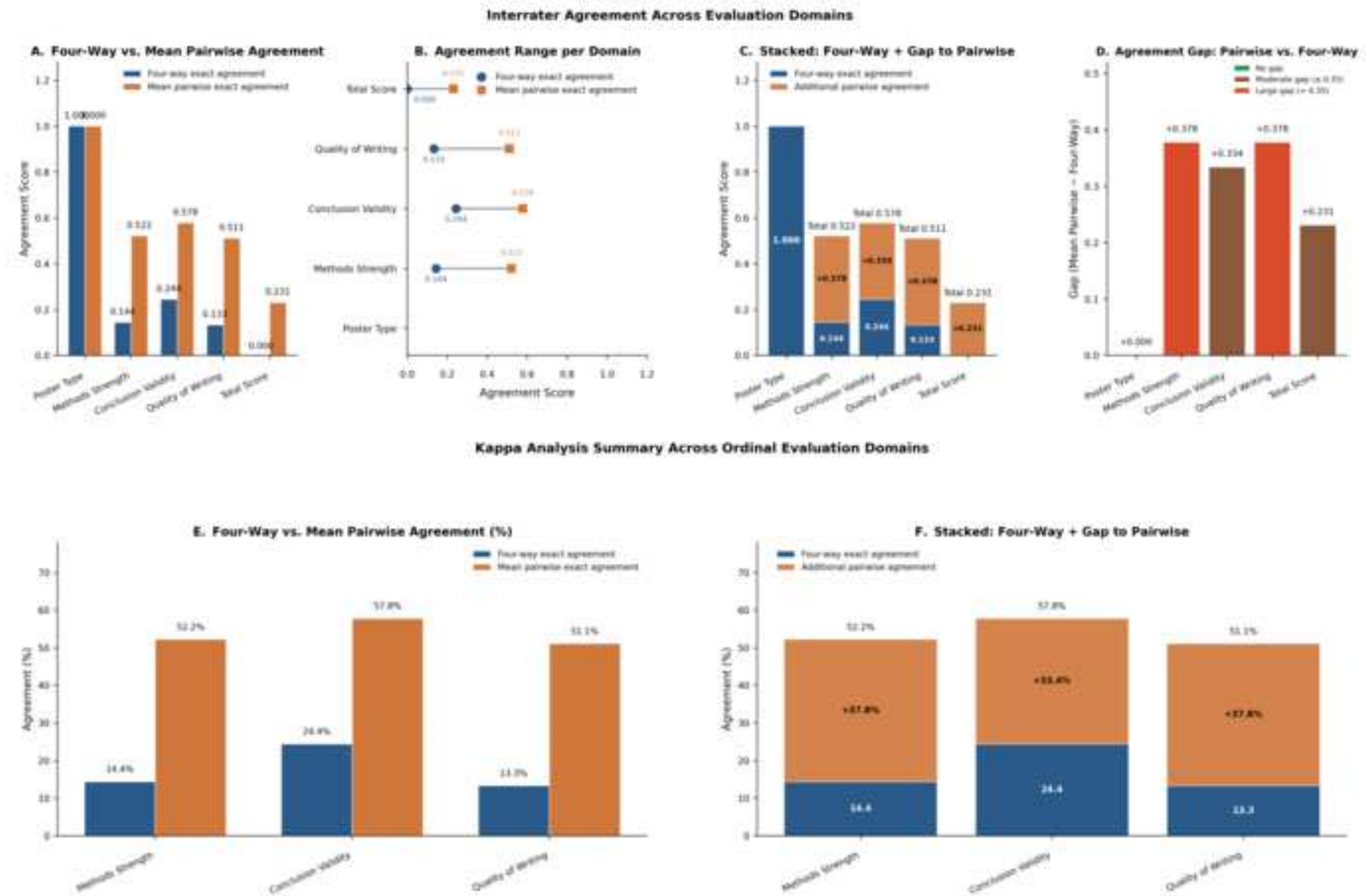

Figure 3

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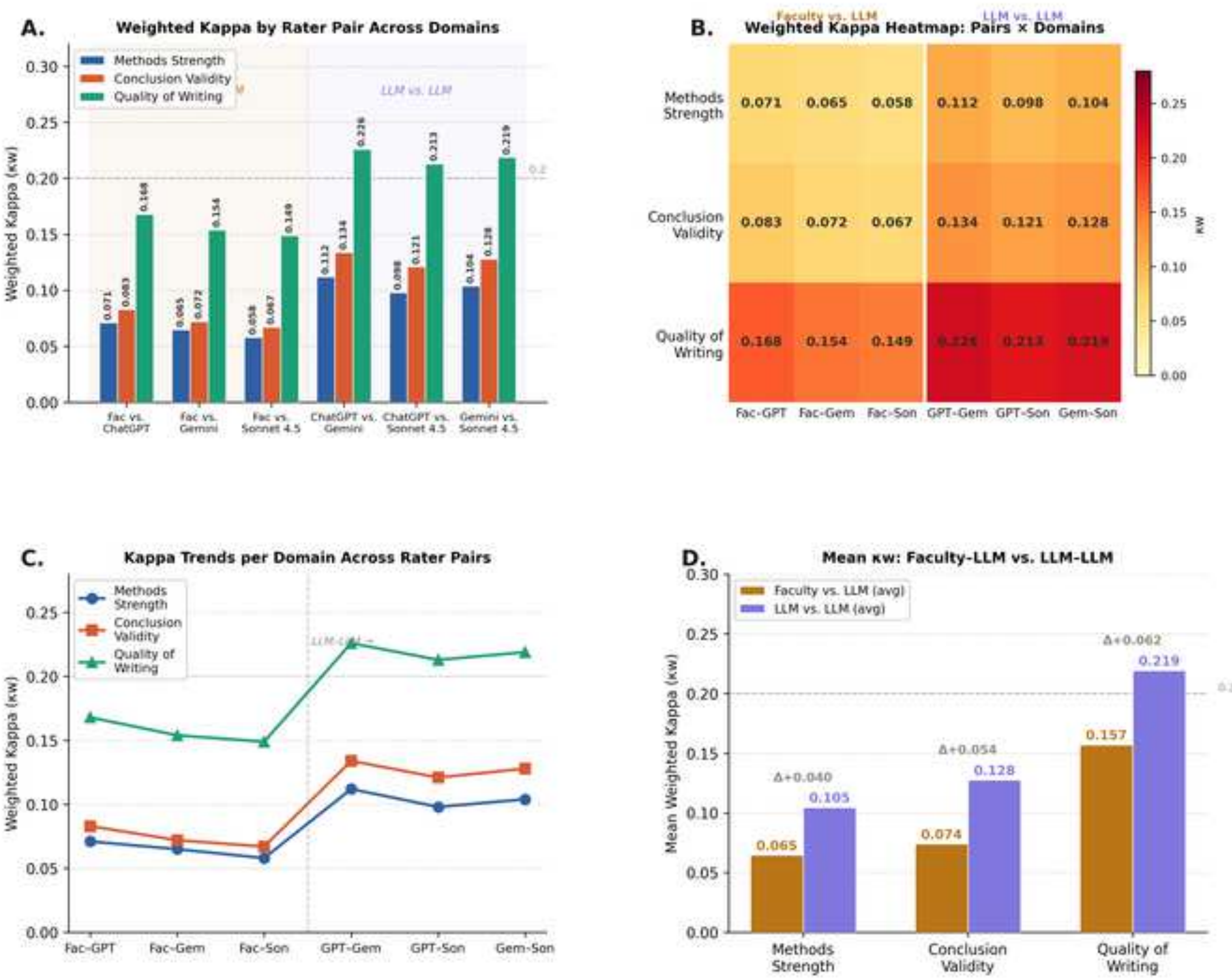
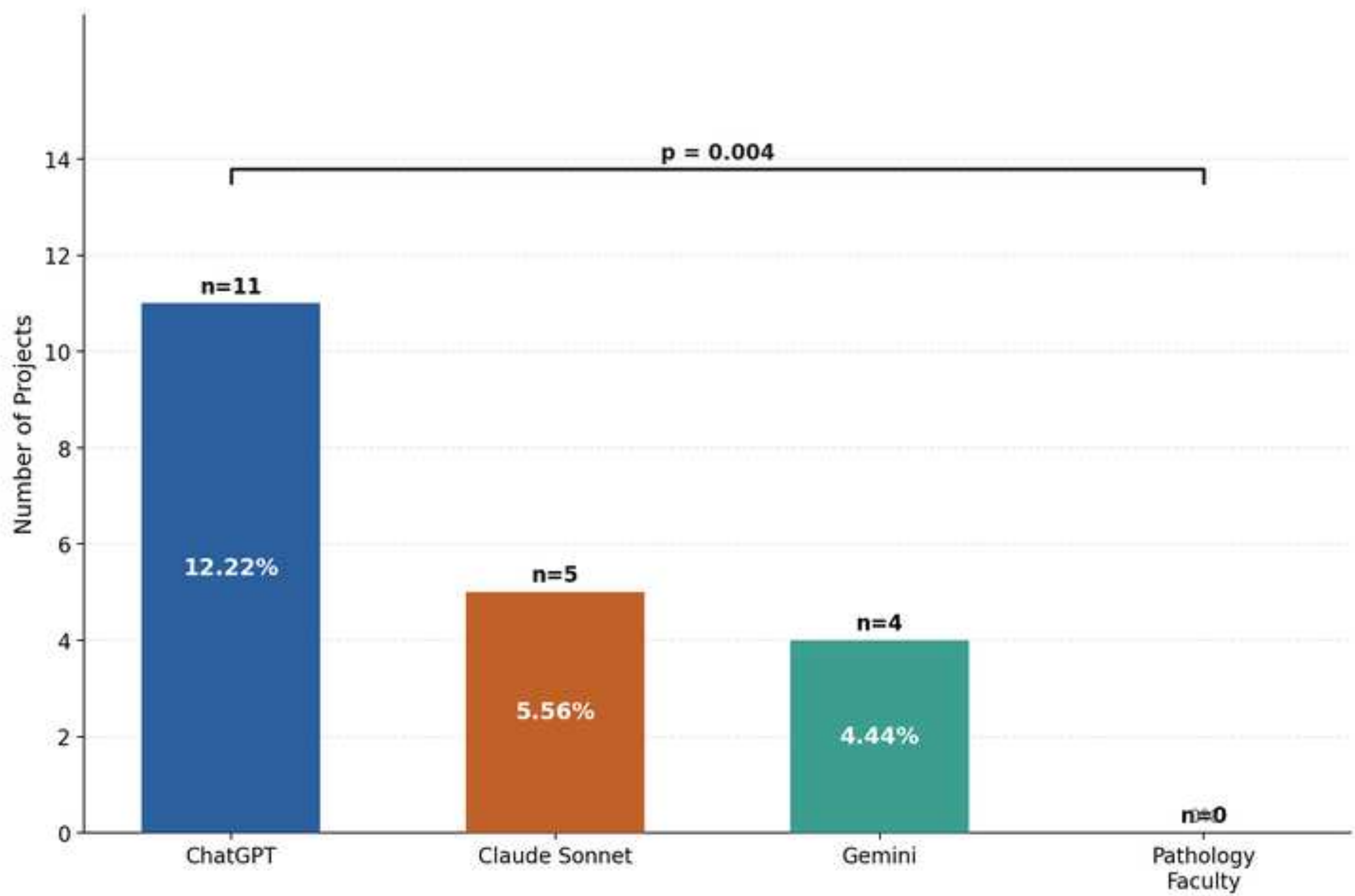
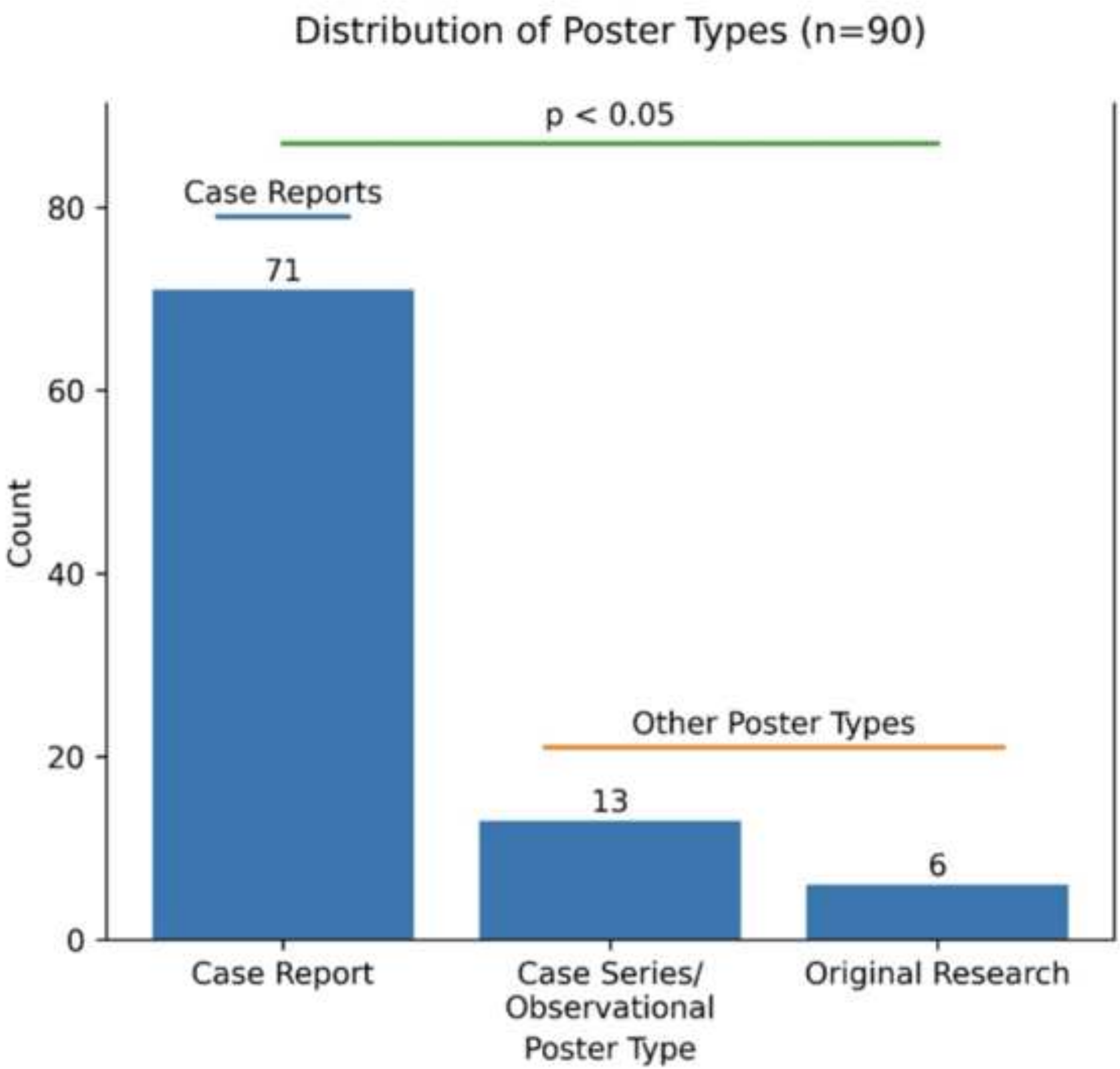


Figure 4




p = 0.004
Number of Projects
0
2
4
6
8
10
12
14
n=11
12.22%
n=5
5.56%
n=4
4.44%
n=0
ChatGPT
Claude Sonnet
Gemini
Pathology Faculty

Supplemental fig 1 



**Supplemental Table 1. System Prompt Used to Instruct Large Language Models for Standardized Evaluation of Pathology Research Posters**

| |
|---|
| *You are an expert academic physician and medical educator with extensive experience reviewing pathology research projects submitted by medical students and residents. Your task is to objectively evaluate the research project using the standardized rubric below: You must strictly follow the scoring criteria and provide numerical scores only within the allowed range. No half points are allowed. Do not invent information that is not present in the text.* |

**Supplemental Table 2. Research Poster Scoring Rubric: Domains of assessment included Poster type, Methods Strength, Conclusion Validity, and Quality of writing.**

| Poster type | Methods Strength | Conclusion Validity | Quality of Writing |
|---|---|---|---|
| • What type of research is this?<br>• Case reports are given a score of 3<br>• Case series (3 or more cases), prospective/retrospective cohort and case/control studies and other types of research is given a score of 4<br>• Original basic research is given as a score of 5. | • Is the study design clearly described, including selection criteria?<br>• Are the measures used reliable?<br>• Are the possible confounding factors addressed?<br>• Are the statistical methods appropriate?<br>Qualitative studies may not have statistical analysis; however, they should still be evaluated based on quality of design description and appropriateness of methods.<br>Case reports should be judged based on the clarity of the case description.<br><u>Scoring guidance:</u><br>• 1- Study design not described. | • Are the conclusions clearly stated and justified by the data?<br>• For case reports, does this presentation increase awareness or improve the workup for an important but rare/unique entity?<br><u>Scoring guidance:</u><br>1 - No conclusions made.<br>2 - Conclusions present but not justified.<br>3 - Conclusions present but weakly supported.<br>4 - Conclusions clearly stated and supported. Knowledge | • Is the writing overall clear, organized, and free of errors? Are the visuals high-quality? Is the layout logical?<br><u>Scoring guidance:</u><br>1- No effort was made in writing and visual composition.<br>2 - Poor writing & visuals with frequent errors in key areas.<br>3 - Average writing and visuals with only minor areas needing improvement.<br>4 - Writing and visuals are above average and well organized. |

| | | | |
|---|---|---|---|
| | • 2- Statistical analysis not done or appropriate to study.<br><br>• 3 - Study design and sampling poorly described.<br><br>• 4 - Possible confounders NOT discussed. Questions regarding methodology. Possible confounders were identified, but there were inadequate controls or statistics. 4 - Fully describe methods. Possible confounders were generally controlled; statistics were appropriate.<br><br>5 - Exemplary method description. No selection bias. Reliable measures, controls, and statistics. | provided is important but NOT likely to change practice.<br><br>5 - Conclusions clearly stated and supported. knowledge likely to change practice | 5 - Writing and visuals are masterfully composed. |

**Supplemental Table 3: Comprehensive Analysis of Interrater Reliability Indices and Estimates.**

| Reliability Index | Poster Type | Methods Strength | Conclusion Validity | Quality of Writing | Total Score: |
|---|---|---|---|---|---|
| **Four-way exact agreement** | 1.000 (perfect agreement) | 0.144 (low agreement) | 0.244 (low agreement) | 0.133 (low agreement) | 0.000 (essentially no exact 4-way agreement) |
| **Mean Pairwise Exact Agreement (6 dyads averaged)** | 1.000 | 0.522 | 0.578 | 0.511 | 0.231 |
| **Krippendorff's α (ordinal)** | 1.000 | 0.037 | 0.013 | 0.062 | 0.125 |
| **Gwet's AC2 (quadratic)** | 1.000 | 0.612 | 0.644 | 0.601 | 0.702 |

**Supplemental Table 4. Overview of Top-Ranked Models Across Projects Cross-Model Consensus on Best Papers Project Models Ranking in Top 3.**

| Project | Models Ranking in Top 3 |
|---|---|
| 65 | Faculty, Sonnet |
| 11 | Faculty, Sonnet, ChatGPT |
| 14 | Sonnet, Gemini |
| 1 | ChatGPT, Gemini |